COMPANY PROFILE

Chengkong Electronics,
Professional data collection product supplier。

More than ten years of ingenious quality assurance
and first-class data collection services



Product Categories

Ⅰ	Analog input module	Ⅱ	Analog output module
Ⅲ	AC input module	Ⅳ	Weighing module
Ⅴ	TC/RTD temperature acquisition module	Ⅵ	Analog input and output module
Ⅶ	Switching/digital module	Ⅷ	Develop custom modules
Ⅸ	signal isolator	Ⅹ	Interface conversion module

Application areas



Automation equipment



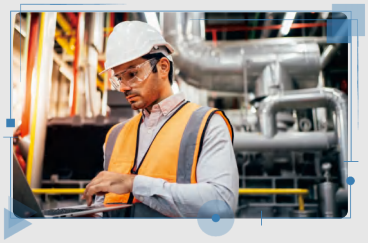
Medical electronics



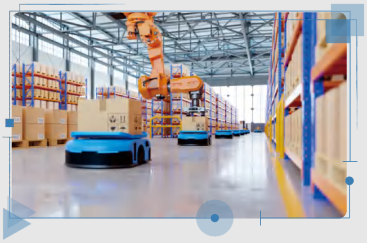
Smart manufacturing



Remote monitoring



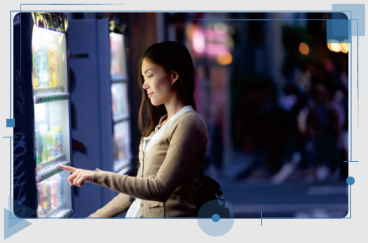
industrial control



Smart warehousing



Instruments anemometer



new retail



8-way switch input and output module

Overview

The DAM module is a new generation of modular data collector based on embedded system. It adopts standard DIN35 rail installation method, which is simple to install on site and flexible to use; it can cope with various field applications. The module is equipped with an isolated RS485 interface, which can communicate with PC or PLC alone, or it can be used in a network with multiple 485 modules.

CK-509 switch quantity/digital quantity/relay input and output data collector can collect up to 8 switch quantity signals (optical coupler input); output 8 digital quantity signals (NPN MOS). Suitable for collecting various IO signals in industrial sites.

CK-509 uses photoelectric isolation technology to effectively ensure reliable and safe data collection.

Technical Parameters

◆Embedded real-time operating system
◆Input and output channels: 8 inputs/8 outputs
◆Input type: optocoupler
◆Output type: NPN mos
◆Wide power supply range: DC 10-30V DC
◆Power supply range: DC 12V or +24V
◆Address/baud rate can be configured by the user
◆Support Modbus RTU protocol, isolated Rs485
◆ESD protection: ±15KV
◆Power consumption: less than 0.5W
◆Isolation withstand voltage: DC 2500V
◆Operating temperature range: -30℃～70℃
◆Industrial grade plastic housing, standard DIN35 rail installation

Application

Automation equipment
Remote monitoring and data collection
Intelligent manufacturing/smart factory
Industrial site control
Smart warehousing and monitoring
Medical and industrial control product development
Packaging and material transfer
Electronic product manufacturing

Function Configuration

model	CK-5161E	CK-508	CK-5009E	CK-509	DAM-5084	DAM-5055
DI (Optocoupler)	16			8	8	5
DO (NPN)			8	8		
Relay Output		8			4	5

CATALOG

1 CK-509 Module Introduction . . . 5
1. 1 Switching data acquisition . . . 5
1.2 Input and output isolation . . . 5
1.4 Surge protection. . . . 5
2 Technical indicators . . . 5
2.1 Switch input. . . . 5
2.2 Switching output . . . 5
3 Port Information . . . 6
3.1 CK-509 Port Arrangement . . . 6
3.2 CK-509 Port Description . . . 6
4 communication . . . 6
4. 1 Communication interface. . . . 6
4.1.1 RS485 Connection. . . . 6
4. 2 Module communication mode . . . 7
4. 2. 1 Master-slave mode . . . 7
4. 3 Communication parameters . . . 7
4.3.1 Mailing address . . . 7
4.4 Communication rate . . . 7
4. 5 Communication Protocol. . . . 7
4.5.1 MODBUS-RTU Protocol . . . 7
4. 5. 2 DAM module MODBUS address allocation . . . 7
4.5.3 DO type output module MODBUS communication example. . . . 8
4.5.4 DI type acquisition module MODBUS communication example . . . 8
6 Electrical parameters . . . 9
6. 1 Module parameters. . . . 9
7 Mechanical specifications . . . 9
7.1 Mechanical Dimensions. . . . 9
8 Installation Method . . . 10
9 Three guarantees and maintenance instructions . . . 10
10 Disclaimer . . . 10
10.1 copyright . . . 10
11 Product display picture . . . 11
12 CK-509 Wiring Diagram . . . 12

CK-509 8-channel switch input and output

Input type: optocoupler, low level valid
Output: RS485 modbus-RTU

CK-509 is a switch quantity/digital quantity/relay input and output data collector, equipped with 8 switch quantity/digital quantity input and output channels. There are also hybrid modules, see selection for details. It is suitable for collecting various switch quantity signals and output switch control signals in industrial sites.



Switching data acquisition

CK-509 adopts advanced data processing technology and can collect various active and passive switch/digital signals in industrial sites. It can meet the industrial sites with high measurement requirements and security, smart buildings, smart homes, power monitoring, process control and other occasions.

Input and output isolation

The product is designed for industrial applications: through DC-DC conversion, the measurement circuit and the main control circuit power supply are isolated; at the same time, the control unit and the signal acquisition unit are electrically isolated using photoelectric isolation technology, effectively ensuring reliable and safe data acquisition.

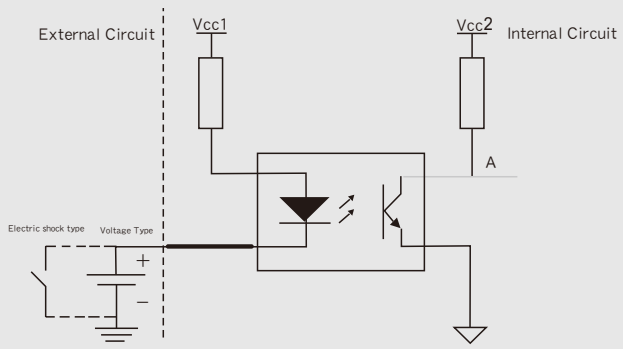
Surge protection

The module is equipped with a transient suppression circuit, which can effectively suppress various surge pulses and protect the module to work reliably in harsh environments.

Technical indicators

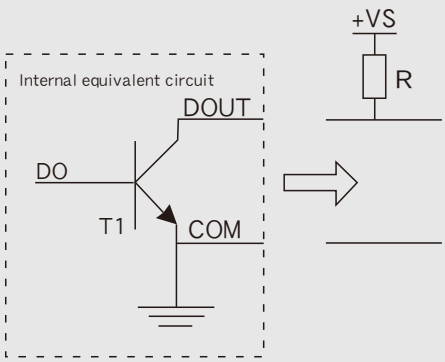
Switch input

◆Number of input channels: 8
◆Input type: optocoupler, low level valid
◆Input equivalent circuit:



Switching output

◆Number of output channels: 8
◆Output type: NPN mos (load capacity: 1W)
◆Output equivalent circuit:



Port Information



CK-509 Port Description

Port	Port ID	Port Function
1	485+	RS485 signal positive input terminal
2	485-	RS485 signal negative input terminal
3	GND	Power Ground
4	VIN	Power input positive terminal
5	DO8	Switching/digital output channels 8
6	DO7	Switching/digital output channels 7
7	DO6	Switching/digital output channels 6
8	DO5	Switching/digital output channels 5
9	DO4	Switching/digital output channels 4
10	DO3	Switching/digital output channels 3
11	DO2	Switching/digital output channels 2
12	DO1	Switching/digital output channels 1
13	IN8	Switching/digital input channels 8
14	IN7	Switching/digital input channels 7
15	IN6	Switching/digital input channels 6
16	IN5	Switching/digital input channels 5
17	IN4	Switching/digital input channels 4
18	IN3	Switching/digital input channels 3
19	IN2	Switching/digital input channels 2
20	IN1	Switching/digital input channels 1

Communication interface

RS485 Connection

The RS485 interface of the CK system module is a standard RS485 interface, which uses differential signal logic. The logic "1" is represented by a voltage difference of +(2~6)V between the two lines; the logic "0" is represented by a voltage difference of -(2~6)V between the two lines. The networking connection of RS485 devices is very simple. You only need to connect the positive and negative ends of the device to the bus. When the communication distance is long, special attention should be paid to the network topology. The RS485 network topology generally adopts a terminal-matched bus structure, and does not support ring or star networks. The lead-out length from the bus to each node should be as short as possible to minimize the impact of the reflected signal in the lead-out line on the bus signal. For more detailed information, please refer to relevant materials.

CK series modules support standard RS485 interface (see the figure below)

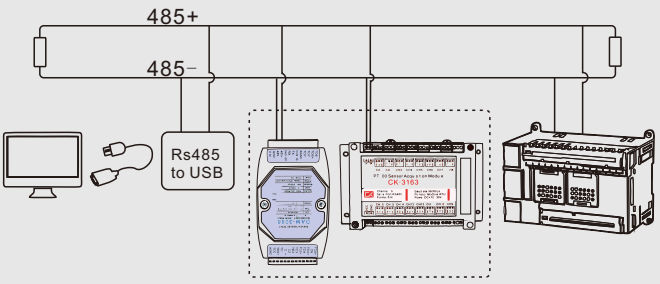
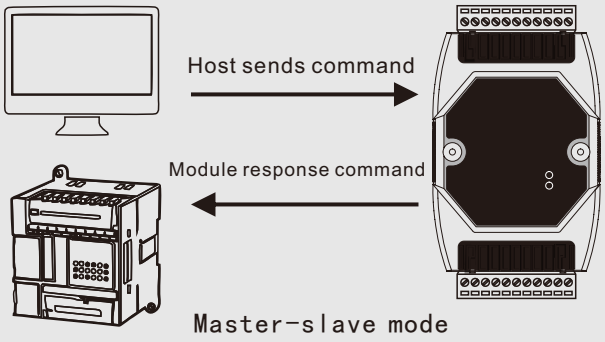


Diagram of CK module connecting to other devices through RS485 interface

Module communication mode

Master-slave mode

The communication mode of CK-509 module is usually master-slave mode (question-answer mode); the host sends commands to the module through the communication interface, and the module responds accordingly after receiving the correct command.



Master-slave mode

Communication parameters (default 9600 8, N, 1 address 01)

Mailing address

The communication address range of the CK-509 module is 01~FA (1~250), and the module address is factory set to 01; the module communication address can be modified by the user through commands according to site needs. For specific methods, please refer to the corresponding commands.

Communication Protocol

Modbus RTU Protocol

Modbus protocol is a universal communication protocol that has been widely used in today's industrial control field. Through this protocol, controllers can communicate with each other or with other devices via a network (such as Ethernet).

The MODBUS address of the DAM module is assigned as follows: (CK-509)

Bit operation register description:

Bit operation function code: 01H (read multi-channel output switch status), 01H (read multi-channel input switch status), 05H (set single-channel switch output status), 0FH (set multi-channel switch output status)

1-8 way switch Output status Corresponding bit address	DO8	DO7	DO6	DO5	DO4	DO3	DO2	DO1	Bit operation function code 01H,05H,0FH	property
	27H	26H	25H	24H	23H	22H	21H	20H		RW
9-16 way switch Output status Corresponding bit address	DO16	DO15	DO14	DO13	DO12	DO11	DO10	DO9	Bit operation function code 01H,05H,0FH	
	2FH	2EH	2DH	2CH	2BH	2AH	29H	28H		RW
Reserved bit address									Bit operation function code	
	37H	36H	35H	34H	33H	32H	31H	30H		
Reserved bit address									Bit operation function code	
	3FH	3EH	3DH	3CH	3BH	3AH	39H	38H		
1-8 way switch Input status Corresponding bit address	DI8	DI7	DI6	DI5	DI4	DI3	DI2	DI1	Bit operation function code 02H	
	47H	46H	45H	44H	43H	42H	41H	40H		R
9-16 way switch Input status Corresponding bit address	DI16	DI15	DI14	DI13	DI12	DI11	DI10	DI9	Bit operation function code 02H	
	4FH	4EH	4DH	4CH	4BH	4AH	49H	48H		R

Communication rate

The CK-509 module RS485 supports baud rates: 1200bps, 2400bps, 4800bps, 9600bps, 14400bps, 19200bps; the module communication rate can be modified by the user through commands according to site needs. For specific methods, please refer to the corresponding commands.

The CK-509 module supports the industrial standard Modbus RTU protocol and can work in the MODBUS slave state. It can communicate with various brands of PLC, RTU or computer. The CK-509 module supports the following MODBUS commands:

Serial number	Command (HEX)	Function	Remark
1	01	Read multiple switch output status (bit)	
2	02	Read multiple switch input status (bit)	
3	03	Read switch status (byte)	
4	05	Set the single-channel switch output status (bit)	
5	06	Write switch output status (byte)	
6	0F	Set the multi-channel switch output status (bit)	

Function code 03H (read), 06H (write)

Command (HEX)	Register address (HEX)	High and low positions	Data Description
03/06	20H	High 8 bits	Switch output status Bit~Bit0 corresponds to channels 16~9
03/06		Lower 8 bits	Switch output status Bit7~Bit0 corresponds to channel 8~1
03	22H	High 8 bits	Switch input status Bit7~Bit0 corresponds to channels 16~9
03		Lower 8 bits	Switch input status Bit7~Bit0 corresponds to channel 8~1

Modbus commands supported by DO output modules:

Serial number	Order(HEX)	Function	Remark
1	01	Read the set DO output status	
1	05	Write the output status of a single DO output channel	
1	0F	Write the output status of one or more DO output channels	Most commonly used

The Modbus address allocation of DO type acquisition modules is as follows:

Order (HEX)	Register address(HEX)	Corresponding PLC address (DEC)	Data Description
01/05/0F	0000	0001	Read and write channel 0 output status (II)
01/05/0F	0001	0002	Read and write channel 1 output status
01/05/0F	0002	0003	Read and write channel 2 output status
01/05/0F	0003	0004	Read and write channel 3 output status

(II) The total number of channels varies depending on the module model.

DO type output module Modbus RTU communication example:

Example	Set DO output status
Module Description	Number of channels: 8, address: 1
Master sends	01 0F 00 00 00 08 01 C3 BE C4
Module Reply	01 0F 00 00 00 08 54 0D
The main station sends analysis	01:Module slave address 0F: Modbus RTUContinuously write coil function code 00 00:0x0000Register start address 00 08:Number of registers written 01:Number of data bytes C3:Output status data, 0xC3 corresponds to binary 0B 11000011. Relay output type: Set state 1 to relay energized. 0C output type: Set state 1 to 0C gate open (connected to GND). BE C4:CRC Check digit
Module reply analysis	01:Module slave address 0F: Modbus RTU Continuously write coil function code 00 00:0x0000 Register start address 00 08:Number of registers written 54 0D:CRC Check digit

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Writing Data	1	1	0	0	0	0	1	1
Channel Number	7	6	5	4	3	2	1	0
Set Status	1	1	0	0	0	0	1	1

Modbus commands supported by DI acquisition modules:

Serial number	Order(HEX)	Function	Remark
1	02	Read digital/switch input	

The Modbus address allocation of DI type acquisition module is as follows:

Order (HEX)	Register address(HEX)	Corresponding PLC address (DEC)	Data Description
02	0000	10001	Digital/switch channel 0 input status (III)
02	0001	10002	Digital/switch channel 1 input status
02	0002	10003	Digital/switch channel 2 input status
02	0003	10004	Digital/switch channel 3 input status

(III) The total number of channels varies depending on the module model.

Modbus RTU communication example of DI type acquisition module:

Example	Read DI input status
Module Description	Number of channels: 8, address: 1
Master sends	01 02 00 00 00 08 79 CC
Module Reply	01 02 01 21 61 90
The main station sends analysis	01:Module slave address 02: Modbus RTU Read input discrete quantity function code 00 00:0x0000 Register start address 00 08:Read register quantity 79 CC:CRCCheck digit
Module reply analysis	01:Module slave address 02: Modbus RTU Read input discrete quantity function code 01:Number of data bytes 21:Input status data, the binary corresponding to 0x21 is 0B00100001. 61 90:CRC Check digit

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Reading Data	0	0	1	0	0	0	0	1
Channel Number	7	6	5	4	3	2	1	0
Channel Status	Low	Low	high	Low	Low	Low	Low	high

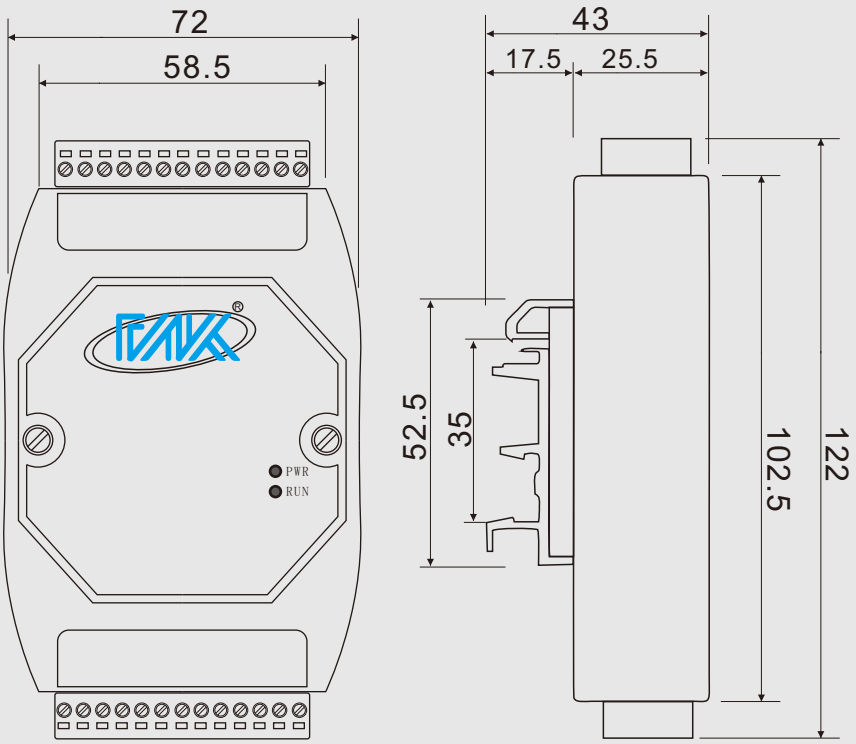
Electrical parameters

Unless otherwise specified, the electrical parameters of the CK-509 data acquisition module are the values when Tamb=25℃.

Module parameters

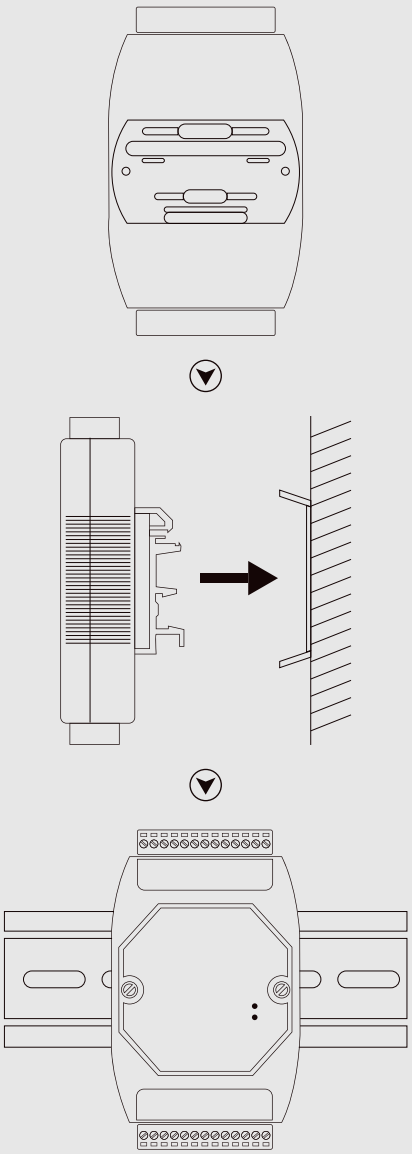
参数	Parameter	最小值 Min	典型值 Typ	最大值 Max	单位 Unit
供电电压	Power Supply	+8	---	+30	V
看门狗复位周期	Watchdog Period		1		S
输入保护	Input Protect		100/60		mA/V

Mechanical Dimensions



Installation method

CK-509 supports DIN35 rail installation. Users can easily install or remove the module on the rail, providing assistance for industrial site application and installation.



Three guarantees and maintenance instructions

Within five years from the date of sale, if the product is damaged or the product quality is lower than the technical indicators under the conditions of storage, transportation and use, the user can return it to the factory for free repair. If the damage is caused by violation of operating regulations and requirements , the device fee and repair fee shall be paid.

Disclaimer

Copyright

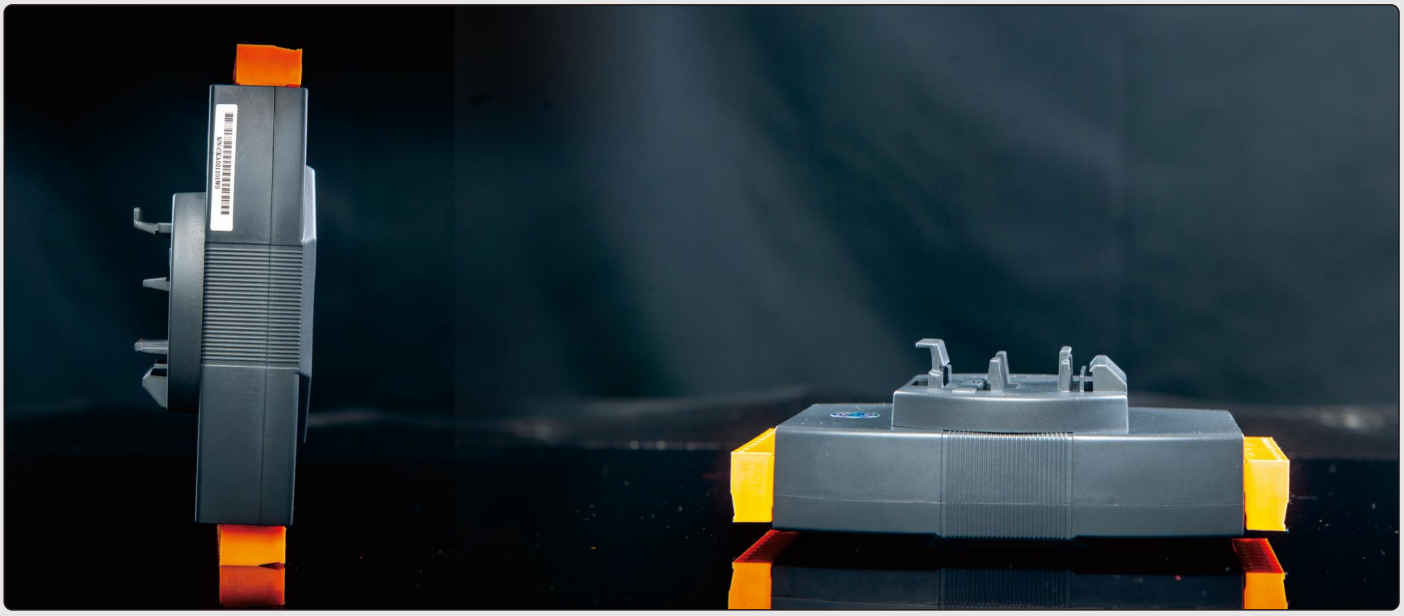
The copyright of the product text and related software described in this manual belongs to Shenzhen Chengkong Electronics Co., Ltd., and its property rights are absolutely protected by national laws. Without the authorization of our company, other companies, units, agents and individuals shall not illegally use and copy them, otherwise the company has the right to impose severe sanctions on national laws.

Product display picture



精工品质 独具匠心







Wiring Diagram

